



DATE: July 31, 2026

FIRST SEMESTER 2026-2027

Course Handout (Part II)

In addition to the part I (General Handout for all courses appended to the timetable), this portion gives further specific details regarding the course –

Course Code: ME G553

Course Title: Moulding Methods for Electronic Components

Instructor-in-Charge: Radha Raman Mishra

1. Course Description

This course provides students with a comprehensive understanding of molding methods used for plastic components in the manufacturing of electronic components. It will introduce students to different molding techniques used for plastic components, and equipment, process considerations, and applications in the electronics industry. The second module provides an overview of the different materials, characterization and testing procedures, and environmental safety considerations in the electronic industry. In the third and fourth modules, students will be exposed to different plastic-moulding processes used in the fabrication of electronic components. The last module will provide basic design aspects related to injection mould, compression mould, transfer mould, and blow mould for fabricating electronic components. Through a strong focus on experimental exposure and industry case studies, students will develop practical insights into advanced manufacturing systems used in the electronics industry.

2. Scope and Objective of the Course

This course introduces theoretical, analytical, and practical knowledge of plastic materials used in the electronic industry, as well as the manufacturing methods for plastic components. It provides students with basic knowledge of advanced casting methods for fabricating plastic components for the electronics industry. It enables learners to understand the principles of various advanced casting methods, the casting machines used, the mechanisms of these processes, the parameters involved, and basic design aspects related to mould. Additionally, this course provides industrial/practical exposure to industrial component development and various application areas with case studies.

3. Course Outcomes

CO No.	Course Outcome
CO1	Develop an understanding of the moulding processes and testing techniques involved in the fabrication of plastic components in the electronics industry.
CO2	Selection of appropriate materials for the electronic industry.
CO3	Design and fabricate different plastic components using moulding processes.
CO4	Acquire basic design skills related to injection mould, compression mould, transfer mould, and blow mould.
CO5	Demonstrate hands-on proficiency and industry readiness by solving practical electronic component fabrication problems through labs, projects, and industry-led sessions.

4. Text Books

- Edwards, P. (2012). *Manufacturing technology in the electronics industry: an introduction*. Springer Science & Business Media. (T1)
- Rosato, D. V. (Ed.). (2012). *Plastics processing data handbook*. Springer Science & Business Media., 2015. (T2)



Please do not print unless necessary





- Nayak, Sanjay K, Pratap C. Padhi, and Y Hidayathullah. Fundamentals of Plastics Mould Design. New Delhi: Tata McGraw Hill Education, 2012. (T3)

5. Reference Books

- Minges, M. L. (1989). *Electronic materials handbook: packaging* (Vol. 1). ASM International. (R1)
- Arabian, J. (2020). Computer integrated electronics manufacturing and testing. CRC Press. (R2)
- Dym, J. B. (1987). Injection molds and molding: a practical manual. Springer Science & Business Media. (R3)
- Rosato, D. V., & Rosato, M. G. (2012). Injection molding handbook. Springer Science & Business Media. (R4)
- Peter Jones, the Mould Design Guide, Smithers Rapra Technology Limited, Shawbury, Shrewsbury, Shropshire, SY4 4NR, UK, 2008. (R5)

6. Course Plan

Module No.	Lecture Session	Reference	Learning outcomes
M1	L (1-2): Introduction to Electronic Components Manufacturing: Overview of the electronic manufacturing industry, types of electronic components manufactured, importance of encapsulation and moulding in electronic components, historical evolution, and trends in electronic moulding, role of mould in protecting and enhancing performance.	T1 and R1	To get exposed to the manufacturing needs of the electronic industry.
M2	L (3-5): Material selection and properties: different metallic, polymeric, ceramic, and composite materials used in the electronics industry. <i>Material Selection Criteria:</i> material properties, including electrical properties, thermal stability, mechanical properties, and environmental resistance. Polymer material characterization and testing.	T1 and R1	To understand the material selection and quality testing for the electronic industry.
M3	L (6-20): Plastic Moulding Techniques: Principles, process fundamentals, tooling involved, material selection, and process parameters for different processes, including injection moulding, compression moulding, transfer moulding, blow moulding, pressure casting, and die casting. Case study.	T2, and R3	To get acquainted with the industrial plastic moulding processes and tooling used.
M4	L (21-35): Design of Moulds: Design aspects of moulds used for plastic component fabrication, Injection Mould Design: different elements, cooling system, ejection system. Design of blow mould, compression mould, and transfer mould. Case study.	T3, R3, R4 and R5	To learn the fundamentals of mould design.
M5	L (35-40): Polymer-based Additive Manufacturing: Introduction, classification, advantages, disadvantages, and applications. Different processes, including Vat photo Polymerization, powder bed fusion, material jetting, material extrusion, and binder jetting: principle, equipment, process, materials, and post-processing.	Class Notes	To understand the fundamentals of precision injection moulding.





7. List of Lab Experiments:

- To understand the injection molding process by producing a simple plastic part of quality based on dimensional accuracy and surface finish.
- To study compression molding techniques by creating a composite electronic housing.
- To study the effect of applied pressure on molten metal flow and solidification behavior during pressure casting.
- To explore transfer molding by encapsulating a small integrated circuit (IC).
- To study the influence of pressure on porosity reduction and the mechanical properties of die-cast components.
- To analyze the effect of injection speed and die temperature on surface finish and dimensional accuracy.
- To study the influence of injection pressure, temperature, and cooling time on part quality and dimensional stability.
- To develop a component using an additive manufacturing process and testing of the part.
- To explore additive manufacturing techniques by creating a mold pattern for sand casting.

8. Evaluation Scheme

Components	Duration (min)	Weightage (%)	Date	Remarks
Mid-Semester Test	90	25	07/10 FN2	CB
Quiz/Seminar/Project	-	20	To be announced in the class	CB/OB
Laboratory	-	20	To be announced in the class	OB
Comprehensive Examination	180	35	08/12 AN	CB

9. **Chamber Consultation Hour:** Thursday, 12 pm- 1 pm.

10. **Notices concerning the course:** All notices concerning the course will be communicated via email/Google Classroom.

11. **Make-up Policy:** Make-up will be permitted only in genuine cases with prior permission. Other than medical reasons, prior permission of the IC is compulsory. No make-up for quizzes.

Instructor-in-Charge

ME G553

